Yoochan Lee

Postdoctoral Researcher

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Research Summary

My research investigates **exploitability** in modern **systems security**. I develop attack techniques and analysis methods that reveal whether and how vulnerabilities can be weaponized in practice, enabling defenders to **prioritize remediation** according to real exploitability rather than bug class alone. My work spans **kernel exploitation**, **side-channel–assisted attacks**, and **data-oriented programming**, with an emphasis on translating offensive insight into principled defensive prioritization.

Academic Appointments

Max Planck Institute for Security and Privacy — Bochum, Germany
Postdoctoral Researcher — *Nov 2025 – Present*
Advisor: Prof. Thorsten Holz

Arizona State University — AZ, USA
Visiting Scholar — *Mar 2024 – Jun 2024*

Education

Seoul National University — Seoul, South Korea
M.S./Ph.D. in Electrical and Computer Engineering — *Sep 2019 – Aug 2025*
Advisor: Prof. Byoungyoung Lee
Samsung Electronics MX Research Scholarship, Sep 2023 – Aug 2025

Hanyang University — Seoul, South Korea
B.S. in Computer Science and Engineering — *Mar 2012 – Feb 2018*

Publications

First-Author Refereed Conference Publications

4. **Heap Localization: Cache Side-Channel based Linux Kernel Heap Exploit Techniques**
   Yoochan Lee, Sihyun Roh, Hyuk Kwon, Byoungyoung Lee, and Thorsten Holz
   *In IEEE Symposium on Security and Privacy (**S&P**), 2026*

3. **DirtyFree: Simplified Data-Oriented Programming in the Linux Kernel**
   Yoochan Lee, Hyuk Kwon, and Thorsten Holz
   *In Network and Distributed System Security Symposium (**NDSS**), Feb 2026*

2. **Pspray: Timing Side-Channel based Linux Kernel Heap Exploitation Technique**
   Yoochan Lee, Jinhan Kwak, Junesoo Kang, Yuseok Jeon, and Byoungyoung Lee
   *In USENIX Security Symposium (**USENIX Security**), Aug 2023*

1. **ExpRace: Exploiting Kernel Races through Raising Interrupts**
   Yoochan Lee, Changwoo Min, and Byoungyoung Lee
   *In USENIX Security Symposium (**USENIX Security**), Aug 2021*

Other Refereed Conference Publications

3. **GHost in the SHELL: A GPU-to-Host Memory Attack and Its Mitigation**
   Sihyun Roh, Woohyuk Choi, Jaeyoung Chung, **Yoochan Lee**, Suhwan Song, and Byoungyoung Lee
   *In IEEE Symposium on Security and Privacy (**S&P**), May 2026*

2. **PeTAL: Ensuring Access Control Integrity against Data-only Attacks on Linux**
   Juhee Kim, Jinbum Park, **Yoochan Lee**, Chengyu Song, Taesoo Kim, and Byoungyoung Lee
   *In ACM Conference on Computer and Communications Security (**CCS**), Oct 2024*

1. **Diagnosing Kernel Concurrency Failures with AITIA**
   Dae R. Jeong, Minkyu Jung, **Yoochan Lee**, Byoungyoung Lee, Insik Shin, and Youngjin Kwon
   *In European Conference on Computer Systems (**EuroSys**), May 2023*

Submitted Manuscripts

- **Know Your Heap Before You Leap: Blind Reconnaissance for Environment-Agnostic Kernel Exploitation**
  Yoochan Lee, and Thorsten Holz
  *Submitted to IEEE Symposium on Security and Privacy (**S&P**), 2027*

- **SlabOrchestra: Environment-Agnostic Heap Manipulation Technique in the Linux Kernel**
  Yoochan Lee, Yonghwa Lee, and Thorsten Holz
  *Submitted to Network and Distributed System Security Symposium (**NDSS**), 2027*

Talks

Invited Academic Talks

- **Revisiting Well-Known Building Blocks for Practical Linux Kernel Exploitation**
  École Polytechnique Fédérale de Lausanne (EPFL), Switzerland, April 2026

Selected Industry and Practitioner Talks

- **Privilege Escalation Exploit using DOP in x86-64 macOS**
  Yoochan Lee, Sangjun Song, Junoh Lee, and Jeongsu Choi
  Hack In The Box Amsterdam 2023

- **Perfect Spray: A Journey From Finding a New Type of Logical Flaw at the Linux Kernel to Developing a New Heap Exploitation Technique**
  Yoochan Lee, Jinhan Kwak, Junesoo Kang, Yuseok Jeon, and Byoungyoung Lee
  Black Hat Europe 2022

- **Exploiting Kernel Races through Taming Thread Interleaving**
  Yoochan Lee, Changwoo Min, and Byoungyoung Lee
  Black Hat USA 2020

Selected Vulnerability Discoveries and Responsible Disclosure

Solidly smart contract: Critical vulnerability enabling unauthorized fund withdrawal.

CVE-2021-31077 (macOS): Kernel heap overflow leading to local privilege escalation.

CVE-2018-4417 (macOS): Kernel information leakage.

CVE-2018-4338 (macOS): Kernel information leakage.

CVE-2018-4084 (macOS): Kernel information leakage.

CVE-2017-7014 (macOS): Arbitrary kernel code execution.

Teaching and Mentoring

Mentoring

Best of the Best (BoB) — Mentor Jul 2023 – Feb 2026
Government-funded cybersecurity education program (South Korea)

Mentored top-tier security students on kernel exploitation and guided responsible vulnerability disclosure (e.g., Linux, NVIDIA).

White Hat School — Lead Mentor Sep 2023 – Dec 2025
Government-funded cybersecurity education program (South Korea)

Designed and delivered a **systems security** curriculum, including lectures and hands-on labs, for beginner-to-intermediate students.

Teaching Assistant

Programming Methodology (430.211) Spring 2025
Seoul National University

Cyber Security and Blockchain (M2177.006300) Spring 2020, Fall 2020, Fall 2021, Fall 2022
Seoul National University

Honors and Awards (Selected)

- **Multiple-time DEFCON CTF Finalist** (2017–2022), Las Vegas, USA
  - **3rd Place**, DEFCON 30 CTF (2022)
  - **4th Place**, DEFCON 29 CTF (2021)
- **1st Place**, Cyber Conflict Exercise and Contest 2018 (GYG), Jeju, South Korea, Oct 2018
- **1st Place**, Secuinside Capture The Bug (Team Minionz), Seoul, South Korea, Jul 2016
- **Top 10**, Best of the Best (BoB), 4th Cohort, Mar 2016

Industry Experience

Raon WhiteHat, Seoul, South Korea
Security Intern: Penetration Testing
Feb 2017 – Aug 2017

Naver Labs, Gyeonggi-do, South Korea
Security Intern: Browser Vulnerability Research (Naver Whale)
Apr 2016 – Jun 2016

ETRI, Daejeon, South Korea
Intern, Network Security Team
Jan 2015 – Feb 2015